MAT 3007 – Optimization

Duality and Interpretations

Lecture 13

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Recap

Recap: Duality

- Given any linear program, we can construct its dual.

	minimize	maximize	
constraints	$\geq b_i$ $\leq b_i$ $= b_i$	≥ 0 ≤ 0 free	variables
variables	≥ 0 ≤ 0 free	$\leq c_j$ $\geq c_j$ $= c_j$	constraints

1. Each primal constraint is associated with a dual variable; each primal variable is associated with a dual constraint.
2. Equality constraints correspond to free variables, vice versa.

Duality Theorems

Weak Duality Theorem

If $\boldsymbol{x}$ and $\boldsymbol{y}$ are feasible points of the primal and dual problem, then:

$$\boldsymbol{b}^\top \boldsymbol{y} \leq \boldsymbol{c}^\top \boldsymbol{x}.$$

- Primal (dual) feasible points can be used to construct bounds for the objective value of the dual (primal) problem.
- If one problem is unbounded, the other one must be infeasible.
- Optimality Conditions: If $\boldsymbol{x}$ is primal feasible, $\boldsymbol{y}$ is dual feasible, and $\boldsymbol{c}^\top \boldsymbol{x} = \boldsymbol{b}^\top \boldsymbol{y}$, then they are both optimal.

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Strong Duality Theorem

If a linear program has an optimal solution, so does its dual, and the optimal value of the primal and dual problems are equal.

Proof for Strong Duality

We prove by using the simplex method. Without loss of generality, we assume the primal problem is in the standard form.

If the primal problem has an optimal solution $\mathbf{x}^*$, then it must be associated with some optimal basis B such that $\mathbf{x}_B = \boldsymbol{A}_B^{-1}\mathbf{b}$ ($\mathbf{x}_B$ is the basic part of $\mathbf{x}^*$). Also, when the simplex method terminates, the reduced costs

$$\mathbf{c}^T - \mathbf{c}_B^T \boldsymbol{A}_B^{-1} A \geq 0 \tag{1}$$

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$$\mathbf{c}^T - \mathbf{c}_B^T \boldsymbol{A}_B^{-1} A \geq 0 \tag{1}$$

Now we define $\mathbf{y}^T = \mathbf{c}_B^T \boldsymbol{A}_B^{-1}$. By above inequality, $\boldsymbol{A}^T\mathbf{y} \leq \mathbf{c}$, i.e., $\mathbf{y}$ is feasible to the dual problem. In addition

$$\mathbf{b}^T\mathbf{y} = \mathbf{c}_B^T \boldsymbol{A}_B^{-1}\mathbf{b} = \mathbf{c}_B^T\mathbf{x}_B = \mathbf{c}^T\mathbf{x}^*$$

Therefore by the weak duality theorem, $\mathbf{y}$ must be optimal to the dual problem and the theorem holds. □

States for Primal and Dual Solutions

The only possible cases for LPs are:

P | D	Finite Optimum	Unbounded	Infeasible
Finite Optimum	✓		
Unbounded			✓
Infeasible		✓	✓

- If both the primal and dual are feasible, then both problems must have optimal solutions. (This is one way to quickly determine if an LP is bounded).
- By strong duality, their optimal values must be the same.

Complementarity Conditions

Complementarity Conditions

Primal		Dual	
min	$\boldsymbol{c}^\top \boldsymbol{x}$	max	$\boldsymbol{b}^\top \boldsymbol{y}$
s.t.	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq \boldsymbol{0}$	s.t.	$\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$

Theorem: Complementarity Conditions

Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be feasible points of the primal and dual problems, respectively. Then, $\boldsymbol{x}$ and $\boldsymbol{y}$ are optimal solutions if and only if:

$$\begin{aligned} x_i > 0 \quad &\Longrightarrow \quad \boldsymbol{A}_i^\top \boldsymbol{y} = c_i \\ \boldsymbol{A}_i^\top \boldsymbol{y} < c_i \quad &\Longrightarrow \quad x_i = 0. \end{aligned}$$

Or in other words, primal variable, dual constraint.

$$x_i \cdot (c_i - \boldsymbol{A}_i^\top \boldsymbol{y}) = 0, \quad \forall\ i.$$

Example

Primal:

$$\begin{array}{rrrrrrr} \text{minimize} & 13x_1 & +10x_2 & +6x_3 & & \\ \text{s.t.} & 5x_1 & +x_2 & +3x_3 & = & 8 \\ & 3x_1 & +x_2 & & = & 3 \\ & x_1, & x_2, & x_3 & \geq & 0 \end{array}$$

$\rightsquigarrow$ Optimal solution: $(1, 0, 1)$.

Dual problem:

$$\begin{array}{rrrrr} \text{maximize} & 8y_1 & +3y_2 & & \\ \text{s.t.} & 5y_1 & +3y_2 & \leq & 13 \\ & y_1 & +y_2 & \leq & 10 \\ & 3y_1 & & \leq & 6 \end{array}$$

$\rightsquigarrow$ Optimal solution: $(2, 1)$.

Verify the complementarity conditions:

$$x_1 \cdot (13 - 5y_1 - 3y_2) = 0,\ x_2 \cdot (10 - y_1 - y_2) = 0,\ x_3 \cdot (6 - 3y_1) = 0.$$

Complementarity in Another Form

Sometimes, we write the dual problem (equivalently) as:

Primal		Dual	
min	$\boldsymbol{c}^\top \boldsymbol{x}$	max	$\boldsymbol{b}^\top \boldsymbol{y}$
s.t.	$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq \boldsymbol{0}$	s.t.	$\boldsymbol{A}^\top \boldsymbol{y} + \boldsymbol{s} = \boldsymbol{c},\ \boldsymbol{s} \geq \boldsymbol{0}$

We call $\boldsymbol{s}$ the dual slack variables. Then, the complementarity conditions can be written as:

$$x_i \cdot s_i = 0 \quad \forall\ i.$$

We sometimes call the complementarity conditions the complementary slackness condition.

General Complementarity Conditions

Primal			Dual		
min	$\boldsymbol{c}^\top \boldsymbol{x}$		max	$\boldsymbol{b}^\top \boldsymbol{y}$	
s.t.	$\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i,$	$i \in M_1,$	s.t.	$y_i \geq 0,$	$i \in M_1$
	$\boldsymbol{a}_i^\top \boldsymbol{x} \leq b_i,$	$i \in M_2,$		$y_i \leq 0,$	$i \in M_2$
	$\boldsymbol{a}_i^\top \boldsymbol{x} = b_i,$	$i \in M_3,$		y_i free,	$i \in M_3$
	$x_j \geq 0,$	$j \in N_1,$		$\boldsymbol{A}_j^\top \boldsymbol{y} \leq c_j,$	$j \in N_1$
	$x_j \leq 0,$	$j \in N_2,$		$\boldsymbol{A}_j^\top \boldsymbol{y} \geq c_j,$	$j \in N_2$
	x_j free,	$j \in N_3,$		$\boldsymbol{A}_j^\top \boldsymbol{y} = c_j,$	$j \in N_3$

Theorem: General Complementarity Conditions

Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be feasible points of the primal and dual problem. Then $\boldsymbol{x}$ and $\boldsymbol{y}$ are optimal if and only if:

$$y_i \cdot (\boldsymbol{a}_i^\top \boldsymbol{x} - b_i) = 0, \quad \forall\, i \quad \text{and} \quad x_j \cdot (\boldsymbol{A}_j^\top \boldsymbol{y} - c_j) = 0, \quad \forall\, j.$$

dual variable — primal slack — primal variable — dual slack variable

Optimality via Complementarity Conditions

Recall that the optimality conditions for LPs were given by:

1. $\boldsymbol{x}$ is primal feasible.
2. $\boldsymbol{y}$ is dual feasible.
3. $\boldsymbol{c}^\top \boldsymbol{x} = \boldsymbol{b}^\top \boldsymbol{y}$.

Now with the complementarity conditions, we can write an equivalent set of conditions:

1. $\boldsymbol{x}$ is primal feasible.
2. $\boldsymbol{y}$ is dual feasible.
3. The complementarity conditions are satisfied.

Solving Problems Via Complementarity Conditions

If we have an optimal solution of the primal problem, then it might be possible to quickly find a dual optimal solution.

Setup: Suppose that the primal problem is in standard form. Let $\boldsymbol{x}$ be the optimal BFS with basis B and $x_i > 0$ for $i \in B$.

Complementarity Conditions: The optimal solution $\boldsymbol{y}$ of the dual problem must satisfy:

$$\boldsymbol{A}_i^\top \boldsymbol{y} = c_i \quad i \in B, \qquad m \text{ constraints.}$$

i.e., $\boldsymbol{y} = (\boldsymbol{A}_B^{-1})^\top \boldsymbol{c}_B$.

Example

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active

active

$$\begin{cases} 5y_1 + 3y_2 = 13 \\ 3y_1 = 6 \end{cases}$$

Because x_1 and x_3 are positive, the first and third constraint should be active for the dual optimal solution. Thus the optimal solution of the dual must be $(2, 1)$.

Dual Solutions in Simplex Tableau

The Dual Optimal Solution and Simplex Tableau

If the initial standard-form problem was formulated by introducing slack variables, the optimal dual solution $(\tilde{\boldsymbol{A}}_B^{-1})^\top \tilde{\boldsymbol{c}}_B$ can be directly obtained from the final simplex tableau.

Primal		Standard Form		Compact	
min	$\boldsymbol{c}^\top \boldsymbol{x}$	min	$\boldsymbol{c}^\top \boldsymbol{x}$	min	$\tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{x}}$
s.t.	$\boldsymbol{A}\boldsymbol{x} \le \boldsymbol{b}$	s.t.	$\boldsymbol{A}\boldsymbol{x} + \boldsymbol{s} = \boldsymbol{b}$	s.t.	$\tilde{\boldsymbol{A}}\tilde{\boldsymbol{x}} = \boldsymbol{b}$
	$\boldsymbol{x} \ge \boldsymbol{0}$		$\boldsymbol{x} \ge \boldsymbol{0}$, $\boldsymbol{s} \ge \boldsymbol{0}$		$\tilde{\boldsymbol{x}} \ge \boldsymbol{0}$

Here, $\tilde{\boldsymbol{x}} = (\boldsymbol{x}; \boldsymbol{s})$, $\tilde{\boldsymbol{c}} = (\boldsymbol{c}; \boldsymbol{0})$ and $\tilde{\boldsymbol{A}} = (\boldsymbol{A}, \boldsymbol{I})$. This compact form has a BFS $(\boldsymbol{0}, \boldsymbol{b})$ (assuming $\boldsymbol{b} \ge \boldsymbol{0}$) and is in canonical form:

$$\begin{array}{|c|c|} \hline \tilde{\boldsymbol{c}}^\top & 0 \\ \hline \tilde{\boldsymbol{A}} & \boldsymbol{b} \\ \hline \end{array} \iff \begin{array}{|c|c|c|} \hline \boldsymbol{c}^\top & \boldsymbol{0}_m & 0 \\ \hline \boldsymbol{A} & \boldsymbol{I}_m & \boldsymbol{b} \\ \hline \end{array}$$

Continue . . .

Suppose after some iterations, the simplex method reaches an optimal solution with basis B. Then the tableau becomes:

$\tilde{\boldsymbol{c}}^\top - \tilde{\boldsymbol{c}}_B^\top \tilde{\boldsymbol{A}}_B^{-1} \tilde{\boldsymbol{A}}$	$-\tilde{\boldsymbol{c}}_B^\top \tilde{\boldsymbol{A}}_B^{-1} \boldsymbol{b}$
$\tilde{\boldsymbol{A}}_B^{-1} \tilde{\boldsymbol{A}}$	$\tilde{\boldsymbol{A}}_B^{-1} \boldsymbol{b}$

$= (c^\top \ 0) - \tilde{c}_B^\top \tilde{A}_B^{-1} (A \ I) = (c^\top - \tilde{c}_B^\top \tilde{A}_B^{-1} A \quad -\tilde{c}_B^\top \tilde{A}_B^{-1})$

$\Longleftrightarrow$

$\boldsymbol{c}^\top - \tilde{\boldsymbol{c}}_B^\top \tilde{\boldsymbol{A}}_B^{-1} \boldsymbol{A}$	$-\tilde{\boldsymbol{c}}_B^\top \tilde{\boldsymbol{A}}_B^{-1}$	$-\tilde{\boldsymbol{c}}_B^\top \tilde{\boldsymbol{A}}_B^{-1} \boldsymbol{b}$
$\tilde{\boldsymbol{A}}_B^{-1} \boldsymbol{A}$	$\tilde{\boldsymbol{A}}_B^{-1}$	$\tilde{\boldsymbol{A}}_B^{-1} \boldsymbol{b}$

Therefore, the final reduced costs corresponding to the original identity matrix part is the optimal dual solution.

dual optimal solution : $(\tilde{A}_B^{-1})^\top \tilde{c}_B$

Example: Production Planning

Consider the production planning problem:

$$
\begin{array}{rrrrrrrl}
\text{minimize} & -x_1 & -2x_2 & & & & & \\
\text{subject to} & x_1 & & +s_1 & & & & = 100 \\
& & 2x_2 & & +s_2 & & & = 200 \\
& x_1 & +x_2 & & & +s_3 & & = 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & & \geq 0
\end{array}
$$

Dual:

$$
\begin{array}{rrrrrl}
\text{maximize} & 100y_1 & +200y_2 & +150y_3 & & \\
\text{subject to} & y_1 & & +y_3 & & \leq -1 \\
& & 2y_2 & +y_3 & & \leq -2 \\
& y_1, & y_2, & y_3 & & \leq 0
\end{array}
$$

Simplex Tableau

The initial tableau for the primal:

B	-1	-2	0	0	0	0
3	1	0	1	0	0	100
4	0	2	0	1	0	200
5	1	1	0	0	1	150

Final tableau:

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

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Final tableau:

B	0	0	0	1/2	1	250
1	1	0	0	-1/2	1	50
3	0	0	1	1/2	-1	50
2	0	1	0	1/2	0	100

What is the dual optimal solution?

- $(y_1, y_2, y_3) = (0, -\frac{1}{2}, -1)$, with objective value -250.

Discussion

If the problem is not derived from adding slack variables , then this deos not work.

- In that case, one can just compute $(\boldsymbol{A}_B^{-1})^\top \boldsymbol{c}_B$.

Interpreting Dual Problems

Interpreting the Dual Problem

We now discuss several examples and provide interpretations of the corresponding dual problems:

- The production planning problem.
- The multi-firm alliance problem.
- ...

Production Planning

Production Planning Problem

	Steel	Iron	Copper	Profit
Alloy 1	1	0	1	\$1
Alloy 2	0	2	1	\$2
Resources	100	200	150	

$$
\begin{aligned}
\text{maximize} \quad & x_1 + 2x_2 \\
\text{subject to} \quad & x_1 \leq 100 \\
& 2x_2 \leq 200 \\
& x_1 + x_2 \leq 150 \\
& x_1, \; x_2 \geq 0.
\end{aligned}
$$

The objective corresponds to profits; the constraints are resource constraints.

Dual of the Production Planning Problem

Write the dual problem:

$$
\begin{array}{rrrrrl}
\text{minimize} & 100p_1 & +200p_2 & +150p_3 & & \\
\text{subject to} & p_1 & & +p_3 & \geq 1 \\
& & 2p_2 & +p_3 & \geq 2 \\
& p_1, & p_2, & p_3 & \geq 0
\end{array}
$$

What is the meaning of this optimization problem?

- Suppose we want to buy all resources from the company at unit prices p_1, p_2 and p_3. What prices should we offer?

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- One unit of resource 1 and resource 3 can produce one unit of product 1 which is worth \$1. Therefore, if $p_1 + p_3 < 1$, the company rather produces by itself (than agree to sell).

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- Suppose we want to buy all resources from the company at unit prices p_1, p_2 and p_3. What prices should we offer?
- One unit of resource 1 and resource 3 can produce one unit of product 1 which is worth \$1. Therefore, if $p_1 + p_3 < 1$, the company rather produces by itself (than agree to sell).
- The buyer wants to buy the resources for the lowest price and the dual problem finds that price.

Discussion

In this example, we interpret the dual variables as fair prices for each resource.

- This is a very common type of interpretation for dual variables.
- Next, we look at an extension of the production planning problem and see how the dual problem can help solving important problem-related questions.

Multi-Firm Alliances

Multi-Firm Alliance Problem

Suppose there is a collection of firms $1, 2, \ldots, m$, each making the same set of products.

- Firm i has access to resources $\boldsymbol{b}_i$ with consumption matrix $\boldsymbol{A}$ and profit $\boldsymbol{c}$ for each product ($\boldsymbol{A}$ and $\boldsymbol{c}$ are the same across all companies).

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$\rightsquigarrow$ Each firm is trying to solve a production problem:

$$\begin{aligned} \text{maximize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}_i \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

We denote the optimal profit for company i by V^i.

Multi-Firm Alliance Problem

Suppose the firms can form an alliance by pooling their resources and produce together.

Let a subset of firms S form an alliance. To maximize the profit of the alliance S, we solve the following LP:

$S \subseteq \{1, 2, \cdots, m\}$

$$\begin{aligned} \text{maximize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \leq \sum_{i \in S} \boldsymbol{b}_i \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

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dual problem

$$\begin{aligned} \text{minimize}_{\boldsymbol{y}} \quad & \left(\sum_{i \in S} \boldsymbol{b}_i^\top\right) \boldsymbol{y} \\ \text{s.t.} \quad & \boldsymbol{A}^\top \boldsymbol{y} \geq \boldsymbol{c} \\ & \boldsymbol{y} \geq \boldsymbol{0} \end{aligned}$$

The optimal profit for the alliance is denoted by V^S.

We call the alliance involving all firms the grand alliance. In a grand alliance, all resources are pooled together. We denote the optimal profit for the grand alliance by V^*.

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One important question before forming an alliance is:

- How to allocate the profit of the alliance to each of its members so that each member/sub-alliance has the incentive to stay in that alliance?

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What constraints do we have?

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1. The total allocation equals the total profit. V^*

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Let us consider the grand alliance. Our question is whether the firms have incentive to form a grand alliance and what allocation rule we should use?

What constraints do we have?

1. The total allocation equals the total profit.
2. The allocated profits to any sub-alliance should be greater than what the sub-alliance could get if they separate from the grand alliance (no subset of players would want to leave the alliance).

Multi-Firm Alliance Problem

In order to make the grand alliance stable, an allocation $(z_1, \ldots, z_m)$ must satisfy:

$$\begin{aligned} \sum_{i=1}^{m} z_i &= V^* \\ \sum_{i \in S} z_i &\geq V^S \qquad \forall\ S \subseteq \{1, \ldots, m\}. \end{aligned}$$

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We call the allocation $(z_1, \ldots, z_m)$ the core of the grand alliance.

- It is not trivial to find such z_i's or to even prove its existence.
- It involves checking 2^m inequalities.

$\rightsquigarrow$ However, we can use duality to find such a solution easily.

Solution: Consider the dual of the grand alliance production problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{y}} \quad & \left(\sum_{i=1}^{m} \boldsymbol{b}_i\right)^\top \boldsymbol{y} \\ \text{s.t.} \quad & \boldsymbol{A}^\top \boldsymbol{y} \geq \boldsymbol{c} \\ & \boldsymbol{y} \geq \boldsymbol{0}. \end{aligned}$$

$\rightsquigarrow$ We denote the optimal solution by $\boldsymbol{y}^*$.

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Theorem: Recovering the Core

Set $z_i = \boldsymbol{b}_i^\top \boldsymbol{y}^*$. Then $(z_1, \ldots, z_m)$ is the core of the grand alliance.

- $\boldsymbol{y}^*$ can be viewed as the prices for the resources and we just allocate according to the available resources of each firm.

Proof

First, by the strong duality theorem, we know that $\sum_{i=1}^{m} z_i = V^*$. Now we only need to show that for any subset S,

$$\sum_{i \in S} z_i \geq V^S.$$

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Consider the production problem for alliance S and its dual:

Primal	$\max_{\mathbf{x}}$	$\mathbf{c}^T\mathbf{x}$	‖	**Dual**	$\min_{\mathbf{y}}$	$\left(\sum_{i \in S} \mathbf{b}_i\right)^T \mathbf{y}$
	s.t.	$A\mathbf{x} \leq \sum_{i \in S} \mathbf{b}_i$	‖		s.t.	$A^T\mathbf{y} \geq \mathbf{c}$
		$\mathbf{x} \geq 0$	‖			$\mathbf{y} \geq 0$

Note that $\mathbf{y}^*$ is still a feasible solution to the dual. Therefore, by the weak duality theorem,

$$\sum_{i \in S} z_i = \left(\sum_{i \in S} \mathbf{b}_i\right)^T \mathbf{y}^* \geq V^S$$

$=$ since the primal problem is maximization

Sensitivity Analysis

Sensitivity Analysis

One important question when studying LP is as follows:

- How do the optimal solution and the optimal value change when the input changes?

This type of problem is called sensitivity analysis.

- We first study this question from a local perspective and then continue with global discussions.

Local Sensitivity

Consider the standard LP:

$$\begin{aligned} \text{minimize}_x \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

We denote the associated optimal value by V.

- If $\boldsymbol{A}$ and $\boldsymbol{c}$ are fixed, V can be viewed as a function of $\boldsymbol{b}$: $V(\boldsymbol{b})$.

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- If $\boldsymbol{A}$ and $\boldsymbol{c}$ are fixed, V can be viewed as a function of $\boldsymbol{b}$: $V(\boldsymbol{b})$.

V is differentiable and

Theorem: Differentiability of the Optimal Value Function

If the dual has a unique optimal solution $\boldsymbol{y}^*$, then $\nabla V(\boldsymbol{b}) = \boldsymbol{y}^*$.

- If the dual optimal solution is not unique (or is unbounded or infeasible), then the gradient is not well-defined.
- If one changes b_i by a small amount Δb_i, then the change of the objective value will be $\Delta b_i y_i^*$

We know that the optimal value V is also the optimal value of the dual problem:

$$\begin{aligned} \text{maximize}_{\boldsymbol{y}} \quad & \boldsymbol{b}^\top \boldsymbol{y} \\ \text{s.t.} \quad & \boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}, \end{aligned}$$

i.e., $V(\boldsymbol{b}) = \boldsymbol{b}^\top \boldsymbol{y}^*$. ← strong duality theorem

$\rightsquigarrow$ If we change $\boldsymbol{b}$ by a small amount $\Delta \boldsymbol{b}$, such that the optimal sol. does not change, then the change of V must be $\Delta \boldsymbol{b}^\top \boldsymbol{y}^*$.

we change $b \to b + \Delta b$.

amount of change in $V(b)$ is $(b+\Delta b)^T y^* - b^T y^* = \Delta b^T y^*$ if the optimal y^* does not change

Similarly, if $\boldsymbol{A}$ and $\boldsymbol{b}$ are fixed, V can be viewed as a function of $\boldsymbol{c}$.

Theorem: Differentiability of $V(\boldsymbol{c})$

If the primal prob. has a unique optimal sol. $\boldsymbol{x}^*$, then $\nabla V(\boldsymbol{c}) = \boldsymbol{x}^*$.

Similarly, if $\boldsymbol{A}$ and $\boldsymbol{b}$ are fixed, V can be viewed as a function of $\boldsymbol{c}$.

Theorem: Differentiability of $V(\boldsymbol{c})$

If the primal prob. has a unique optimal sol. $\boldsymbol{x}^*$, then $\nabla V(\boldsymbol{c}) = \boldsymbol{x}^*$.

If one changes c_i by a small amount Δc_i, then the change of the objective value will be $\Delta c_i x_i^*$.

- $\rightsquigarrow$ **Reason:** If we change $\boldsymbol{c}$ by a small amount $\Delta \boldsymbol{c}$, such that the optimal solution does not change, then the change of V must be $\Delta \boldsymbol{c}^\top \boldsymbol{x}^*$.

$c \to c + \Delta c$

$(c+\Delta c)^T x^* - c^T x^* = \Delta c^T x^*$

The latter results also hold for inequality constraints (or maximization problems):

$$\begin{aligned} \text{maximize}_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$

We have:

1. If the dual has a unique optimal solution $\boldsymbol{y}^*$, then $\nabla V(\boldsymbol{b}) = \boldsymbol{y}^*$.
2. If the primal has a unique optimal solution $\boldsymbol{x}^*$, then $\nabla V(\boldsymbol{c}) = \boldsymbol{x}^*$.

- To see why this must be true, one can add a slack variable and transform it back to the standard form. We can then use the earlier result.

Example: Production Planning

$$
\begin{array}{rrrrl}
\text{maximize} & x_1 & +2x_2 & & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.

The dual problem is

$$
\begin{array}{rrrrrl}
\text{minimize} & 100y_1 & +200y_2 & +150y_3 & & \\
\text{subject to} & y_1 & & & +y_3 & \geq 1 \\
& & 2y_2 & & +y_3 & \geq 2 \\
& y_1, & y_2, & & y_3 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$ with optimal value 250.

Example: Continued — I

$$
\begin{array}{rrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Example: Continued — I

$c = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
 & & 2x_2 & \leq 200 \\
 & x_1 & +x_2 & \leq 150 \\
 & x_1, & x_2 & \geq 0
\end{array}
$$

$b = \begin{pmatrix} 100 \\ 200 \\ 150 \end{pmatrix}$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Let assume $\Delta b_2 = 2$ is "small enough" for now.

Q1: What is the optimal value if we have 202 units of resource 2?

b_2 200 → 202

- It will change by $\Delta b_2 y_2^* = 1$. Therefore, the optimal value would be 251 ($\rightsquigarrow$ check with CVX: ✓).

$\Delta b_2 = 2$

$\Delta b_2 \cdot y_2^* = 2 \times \frac{1}{2} = 1$

Example: Continued — II

$$
\begin{array}{rrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Example: Continued — II

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

b_1 100 → 99

assume $\Delta b_1 = -1$ is "small enough" --

$\Delta b_1 = -1$

Q2: What is the optimal value if we have 99 units of resource 1?

$\Delta b_1 \cdot y_1^* = -1 \times 0 = 0$

- It will change by $\Delta b_1 y_1^* = 0$. Therefore, the optimal value would be unchanged ($\rightsquigarrow$ check with CVX: ✓).

Example: Continued — III

$$
\begin{array}{rrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Example: Continued — III

$c = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$\begin{array}{lrrl} \text{maximize} & x_1 & +2x_2 & \\ \text{subject to} & x_1 & & \leq 100 \\ & & 2x_2 & \leq 200 \\ & x_1 & +x_2 & \leq 150 \\ & x_1, & x_2 & \geq 0 \end{array}$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

c_1 $1 \to 1.02$

$\Delta c_1 = 0.02$.

Q3: What's the opt. value if the profit of product 1 becomes 1.02?

$\Delta c_1 \cdot x_1^* = 0.02 \times 50 = 1$

- It will increase by $\Delta c_1 x_1^* = 1$. Therefore, the optimal value would be 251 ($\rightsquigarrow$ check with CVX: ✓).

Example: Continued - IV

$$
\begin{array}{rrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Example: Continued - IV

$$
\begin{array}{lrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
 & & 2x_2 & \leq 200 \\
 & x_1 & +x_2 & \leq 150 \\
 & x_1, & x_2 & \geq 0
\end{array}
$$

$c_2: 2 - 1.97$

$\Delta c_2 = -0.03.$

$\Delta c_2 \times x_2^* = -0.03 \times 100 = -3$

The optimal solution is $\boldsymbol{x}^* = (50, 100)$ with optimal value 250.
The dual optimal solution is $\boldsymbol{y}^* = (0, 0.5, 1)$.

Q4: What's the opt. value if the profit of product 2 becomes 1.97?

- It will decrease by $\Delta c_2 x_2^* = -3$. Therefore, the optimal value would be 247 ($\rightsquigarrow$ check with CVX: ✓).

Today's lecture

- Interpretation of Dual Problems
- Intro to Sensitivity Analysis

Next lecture

- Sensitivity Analysis

Questions?